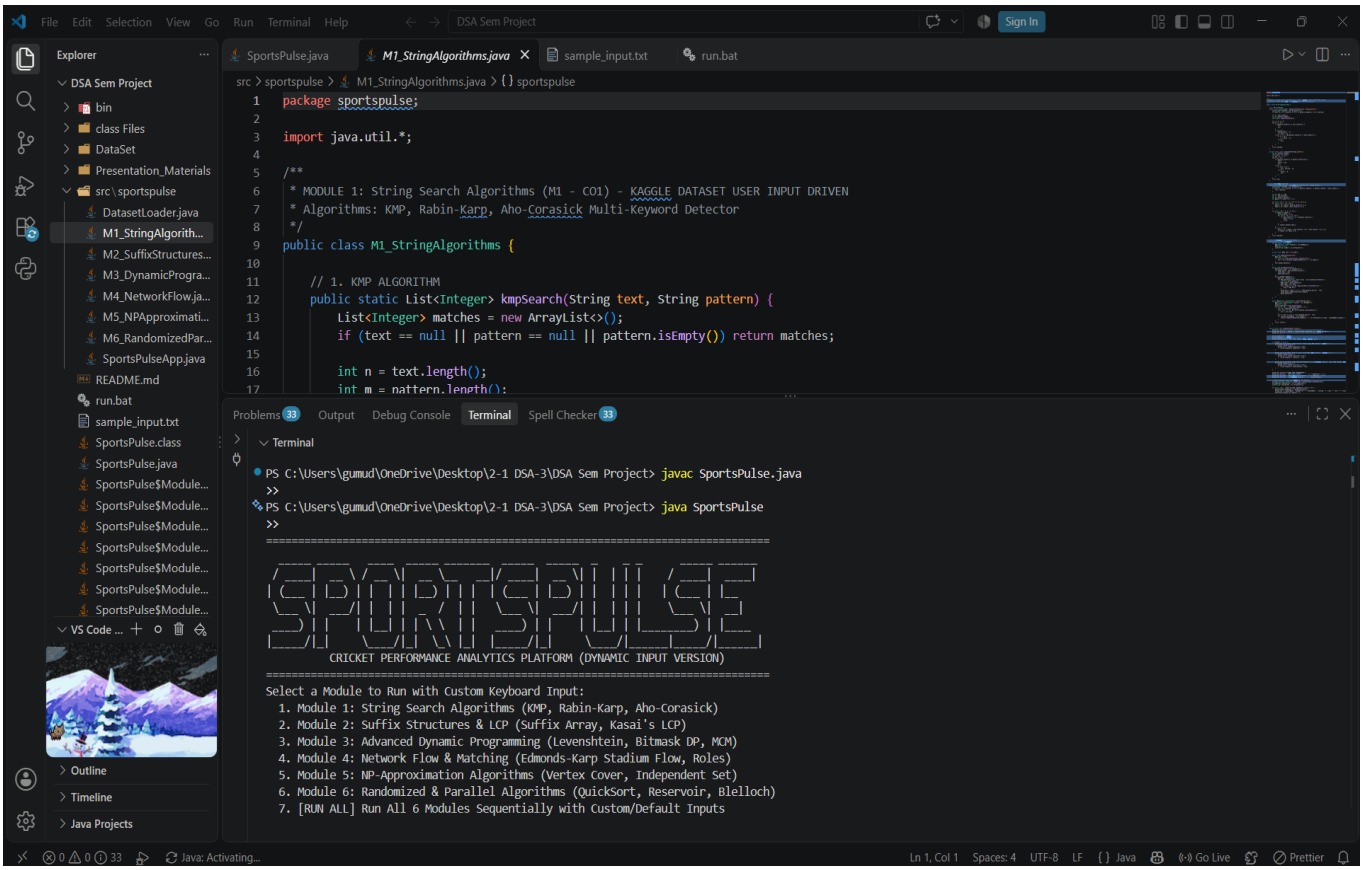
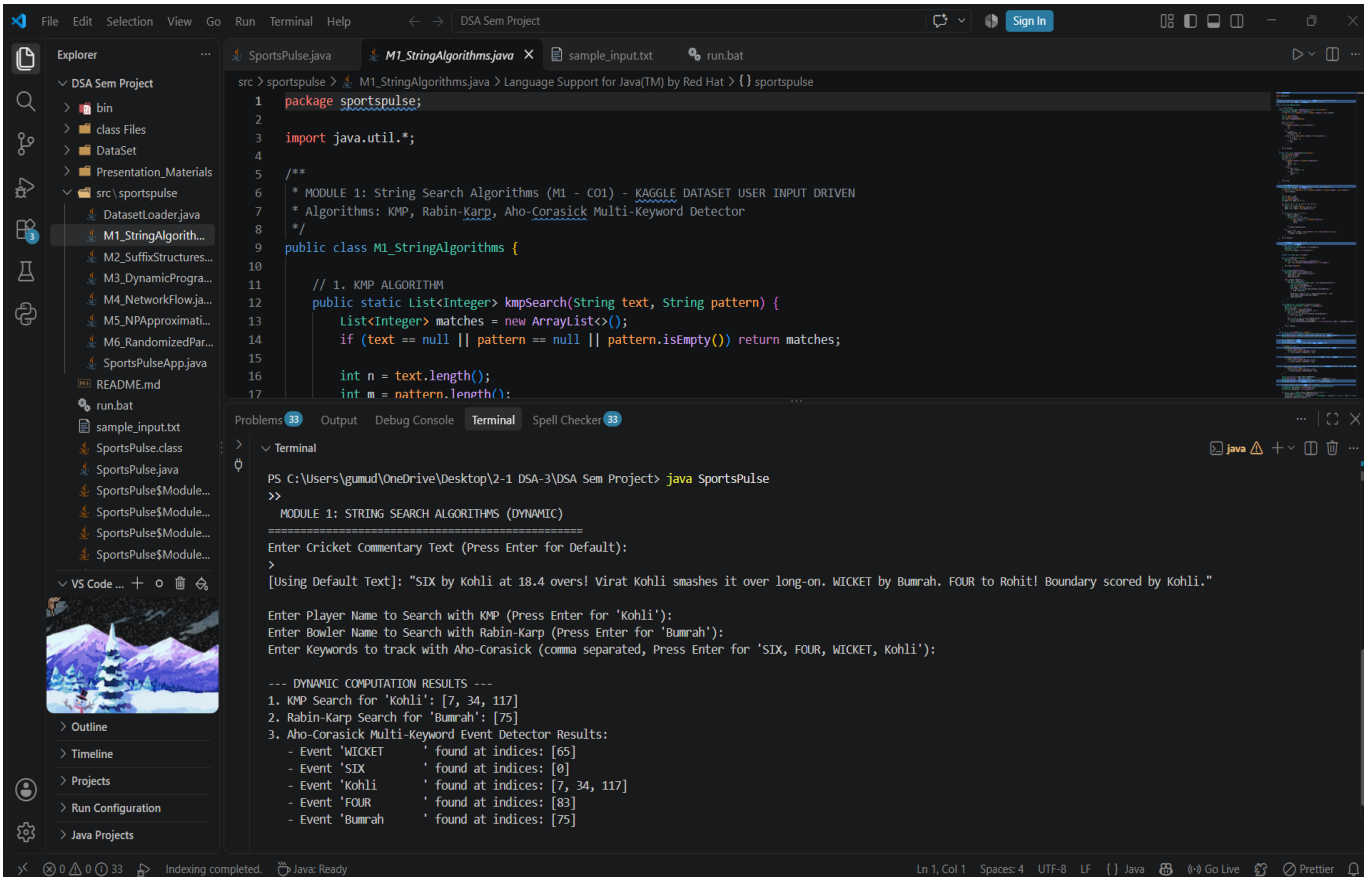
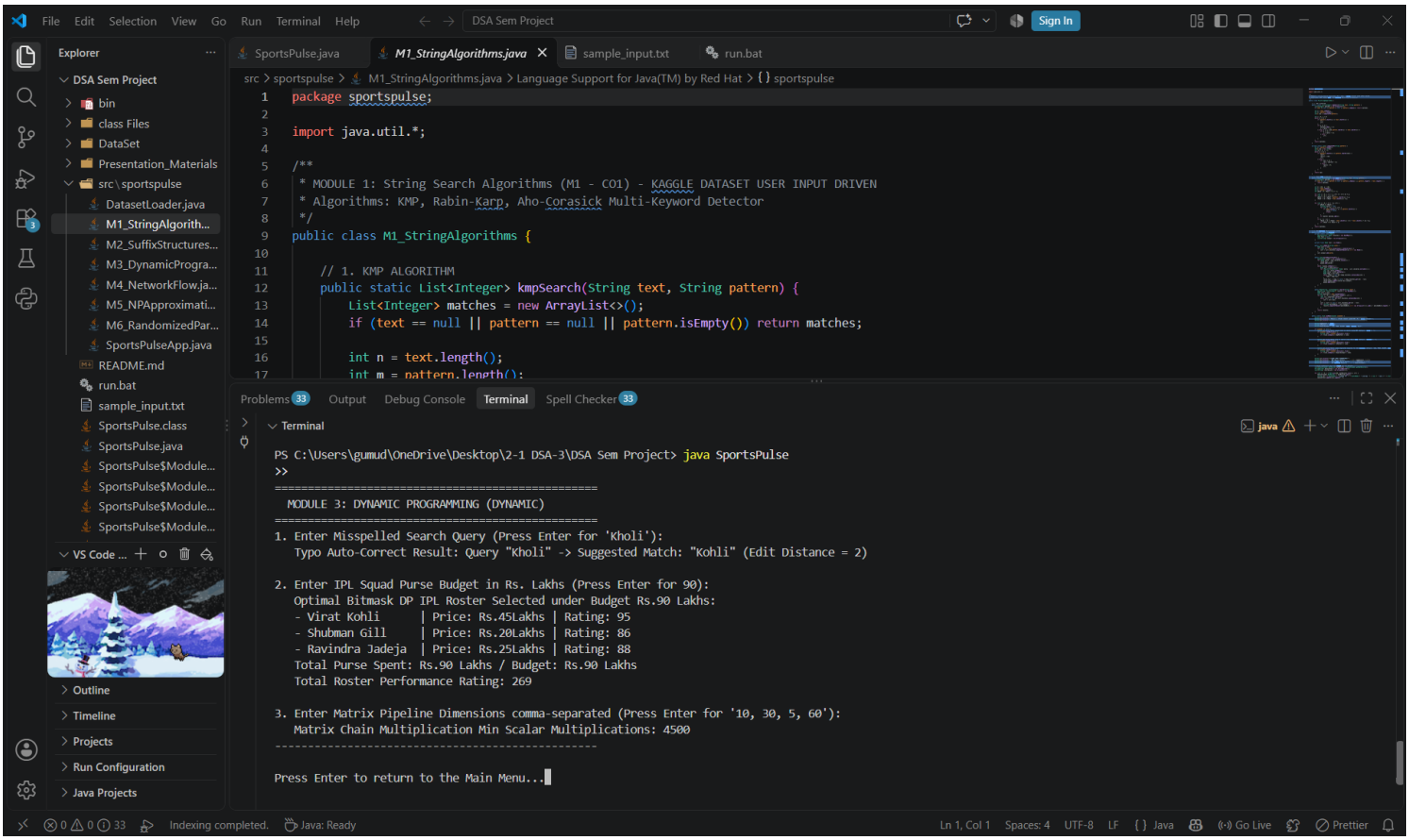
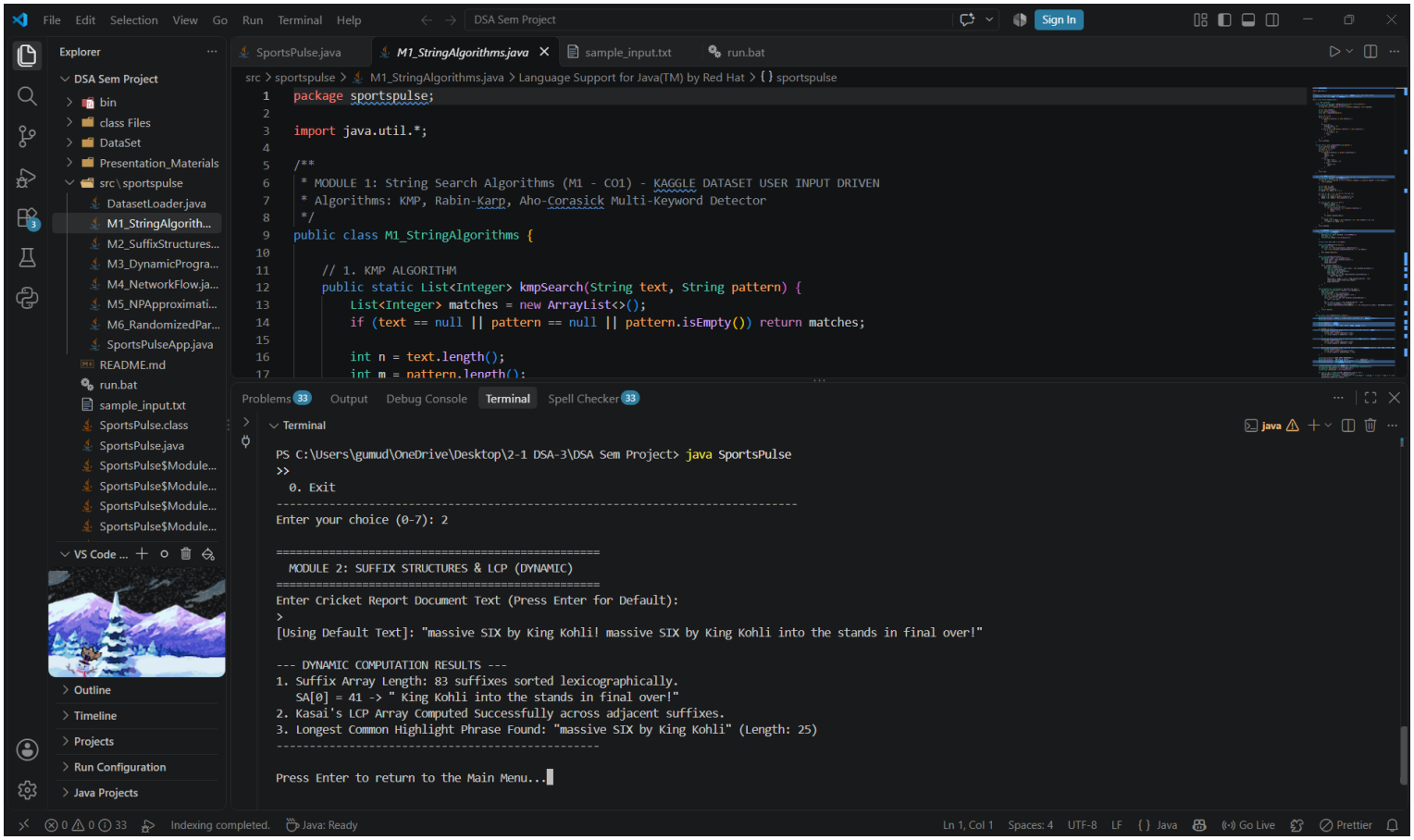


Module 1 : Output



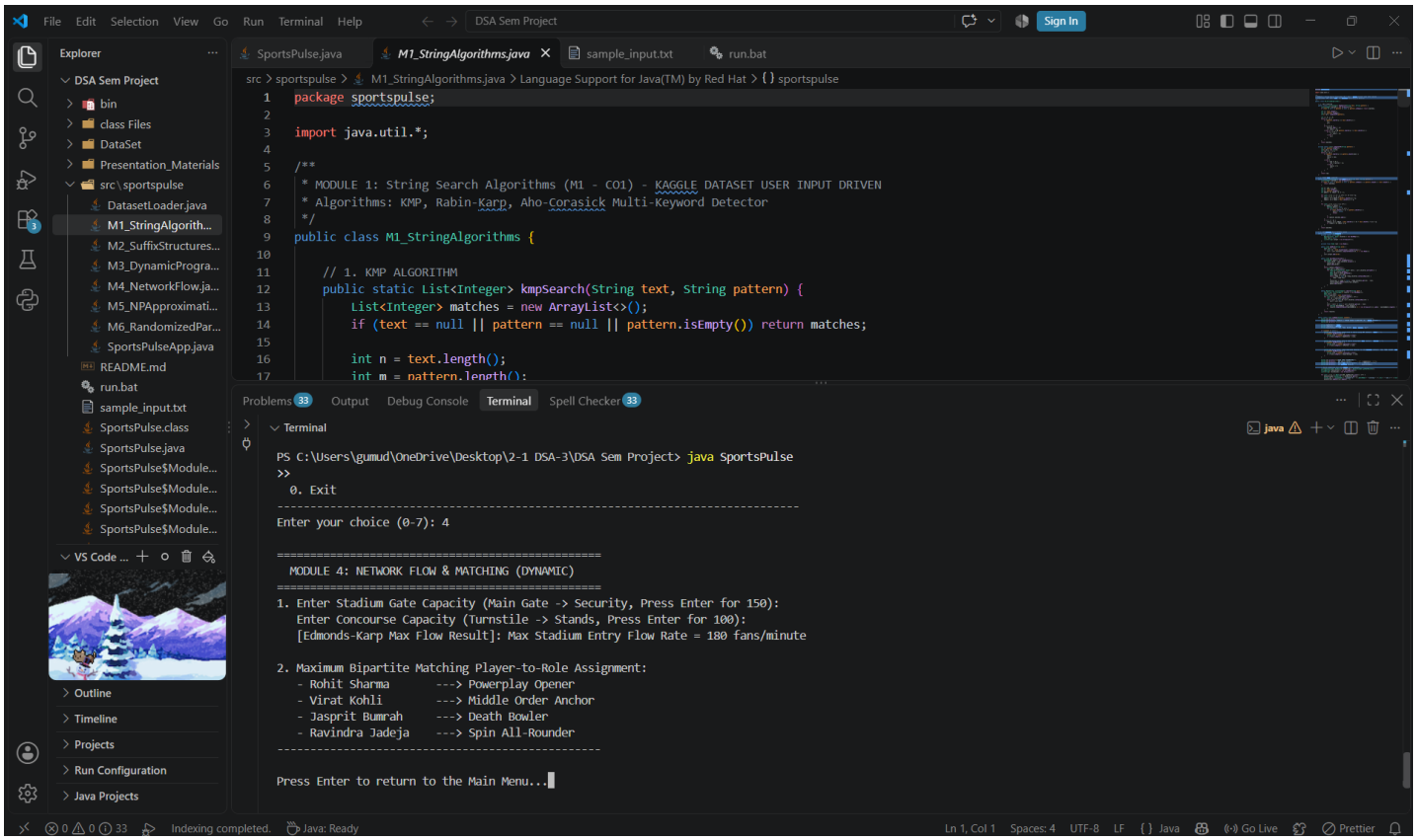

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M4_NetworkFlow.ja...
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M6_RandomizedPar...
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15
16 int n = text.length();
17 int m = pattern.length();
Problems 33
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Spell Checker 33
Terminal
java
PS C:\Users\gumud\OneDrive\Desktop\2-1 DSA-3\DSA Sem Project> java SportsPulse
>>
==========
MODULE 3: DYNAMIC PROGRAMMING (DYNAMIC)
==========
1. Enter Misspelled Search Query (Press Enter for 'Kholi'):
Typo Auto-Correct Result: Query "Kholi" -> Suggested Match: "Kohli" (Edit Distance = 2)
2. Enter IPL Squad Purse Budget in Rs. Lakhs (Press Enter for 90):
Optimal Bitmask DP IPL Roster Selected under Budget Rs.90 Lakhs:
- Virat Kohli | Price: Rs.45Lakhs | Rating: 95
- Shubman Gill | Price: Rs.20Lakhs | Rating: 86
- Ravindra Jadeja | Price: Rs.25Lakhs | Rating: 88
Total Purse Spent: Rs.90 Lakhs / Budget: Rs.90 Lakhs
Total Roster Performance Rating: 269
3. Enter Matrix Pipeline Dimensions comma-separated (Press Enter for '10, 30, 5, 60'):
Matrix Chain Multiplication Min Scalar Multiplications: 4500

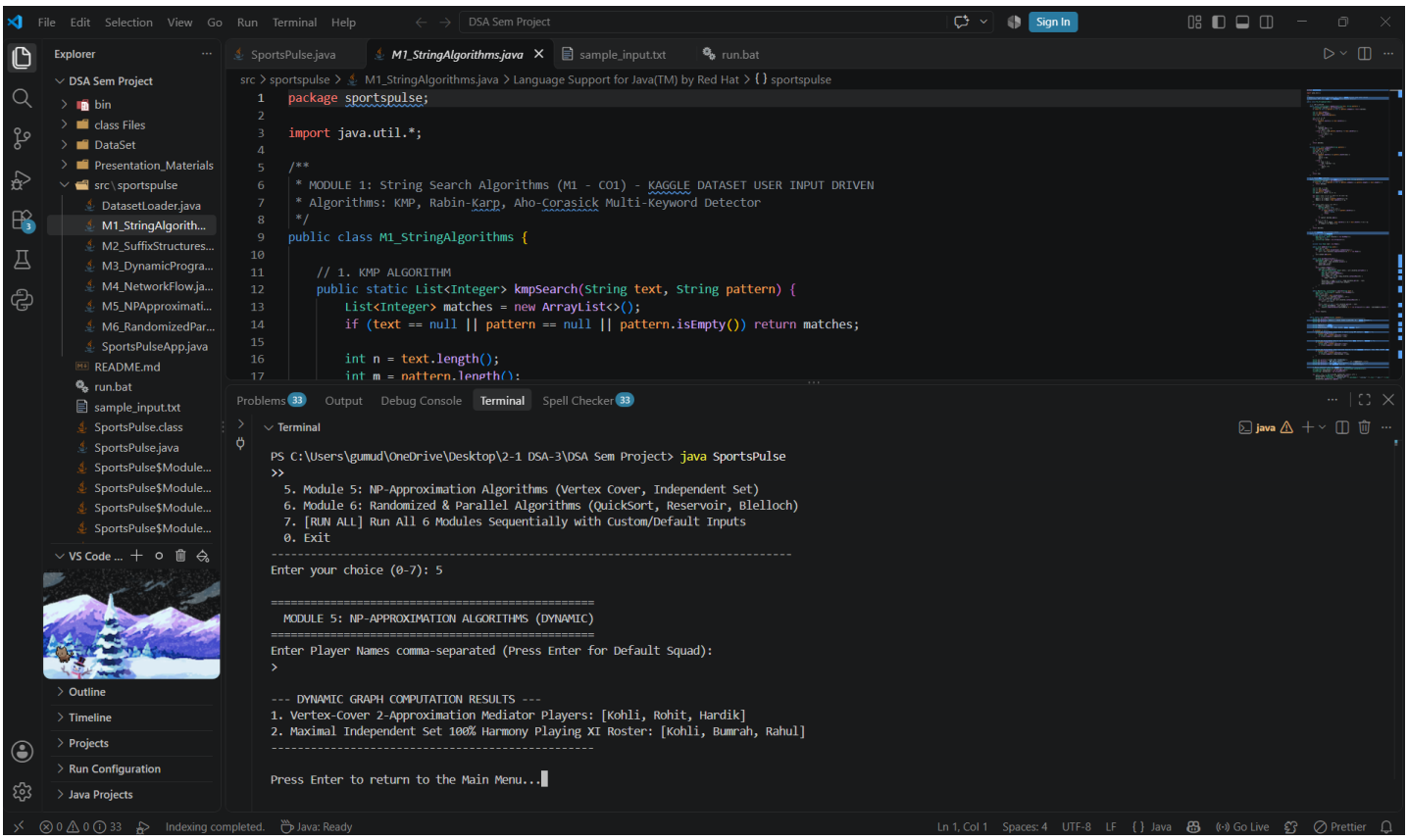
Press Enter to return to the Main Menu...
Indexing completed.
Java: Ready
Ln 1, Col 1
Spaces: 4
UTF-8
LF
{ } Java
Go Live
Prettier

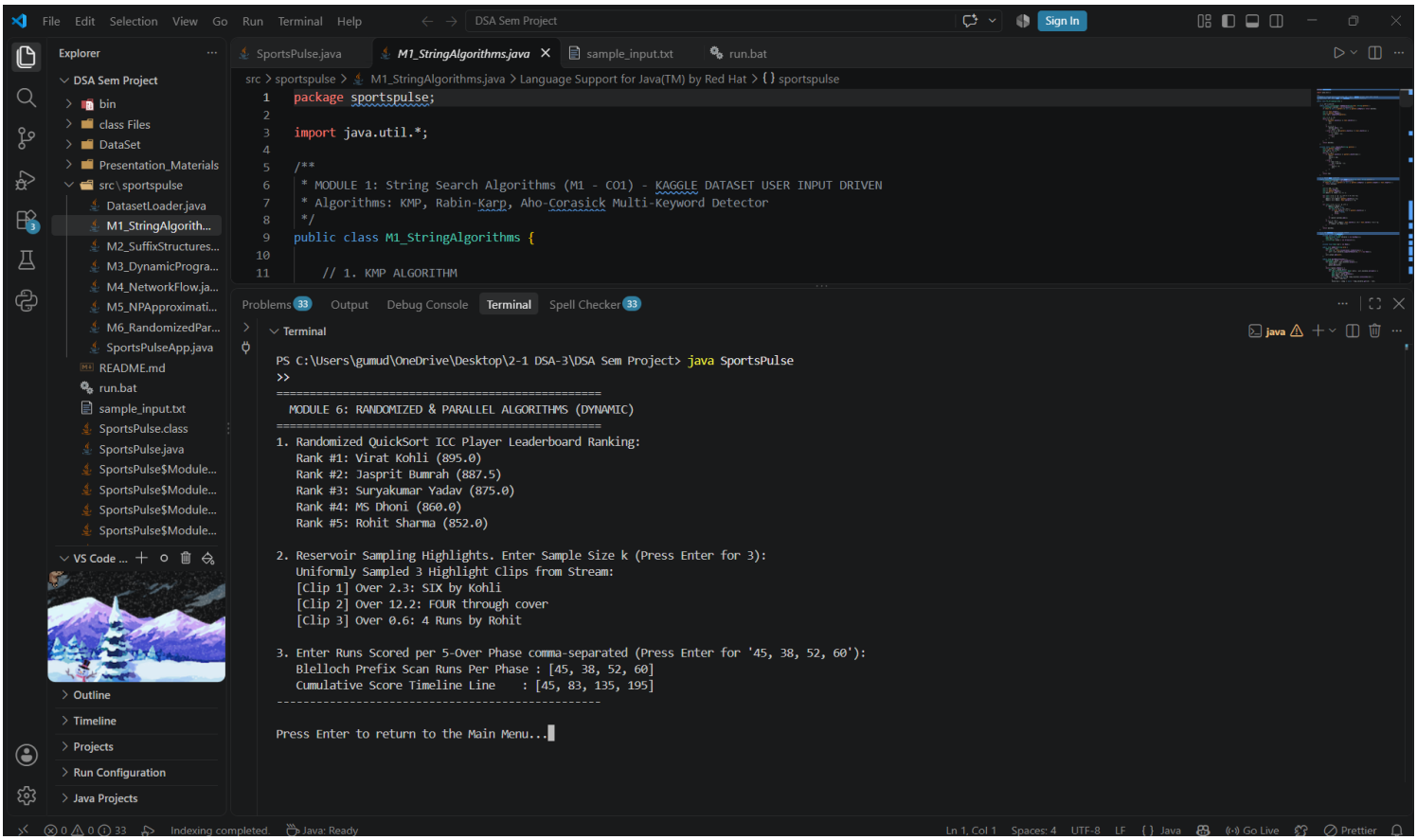

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Terminal
Spell Checker 33
Terminal
java
PS C:\Users\gumud\OneDrive\Desktop\2-1 DSA-3\DSA Sem Project> java SportsPulse
>>
0. Exit

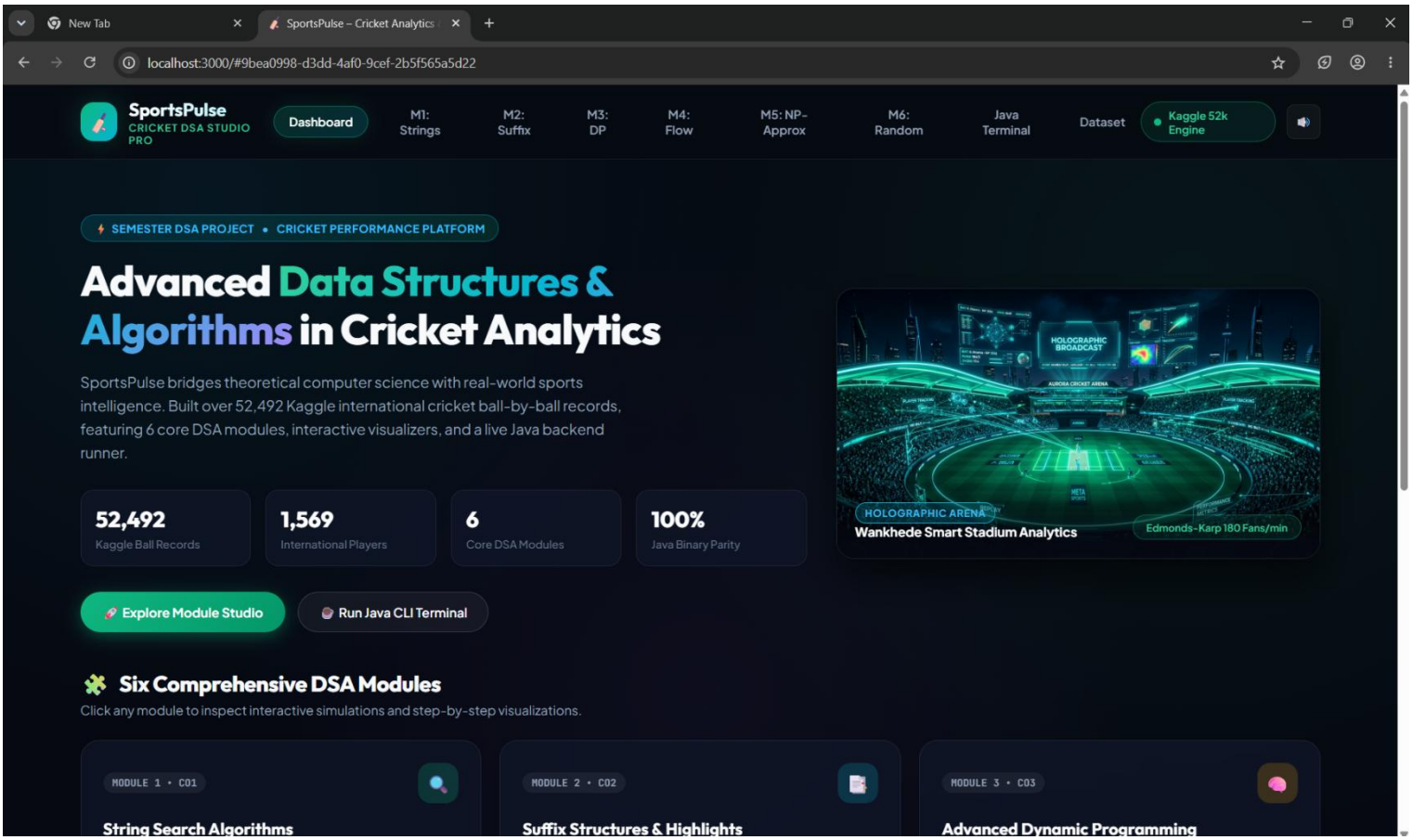
Enter your choice (0-7): 2
==========
MODULE 2: SUFFIX STRUCTURES & LCP (DYNAMIC)
==========
Enter Cricket Report Document Text (Press Enter for Default):
>
[Using Default Text]: "massive SIX by King Kohli! massive SIX by King Kohli into the stands in final over!"
--- DYNAMIC COMPUTATION RESULTS ---
1. Suffix Array Length: 83 suffixes sorted lexicographically.
SA[0] = 41 -> " King Kohli into the stands in final over!"
2. Kasai's LCP Array Computed Successfully across adjacent suffixes.
3. Longest Common Highlight Phrase Found: "massive SIX by King Kohli" (Length: 25)

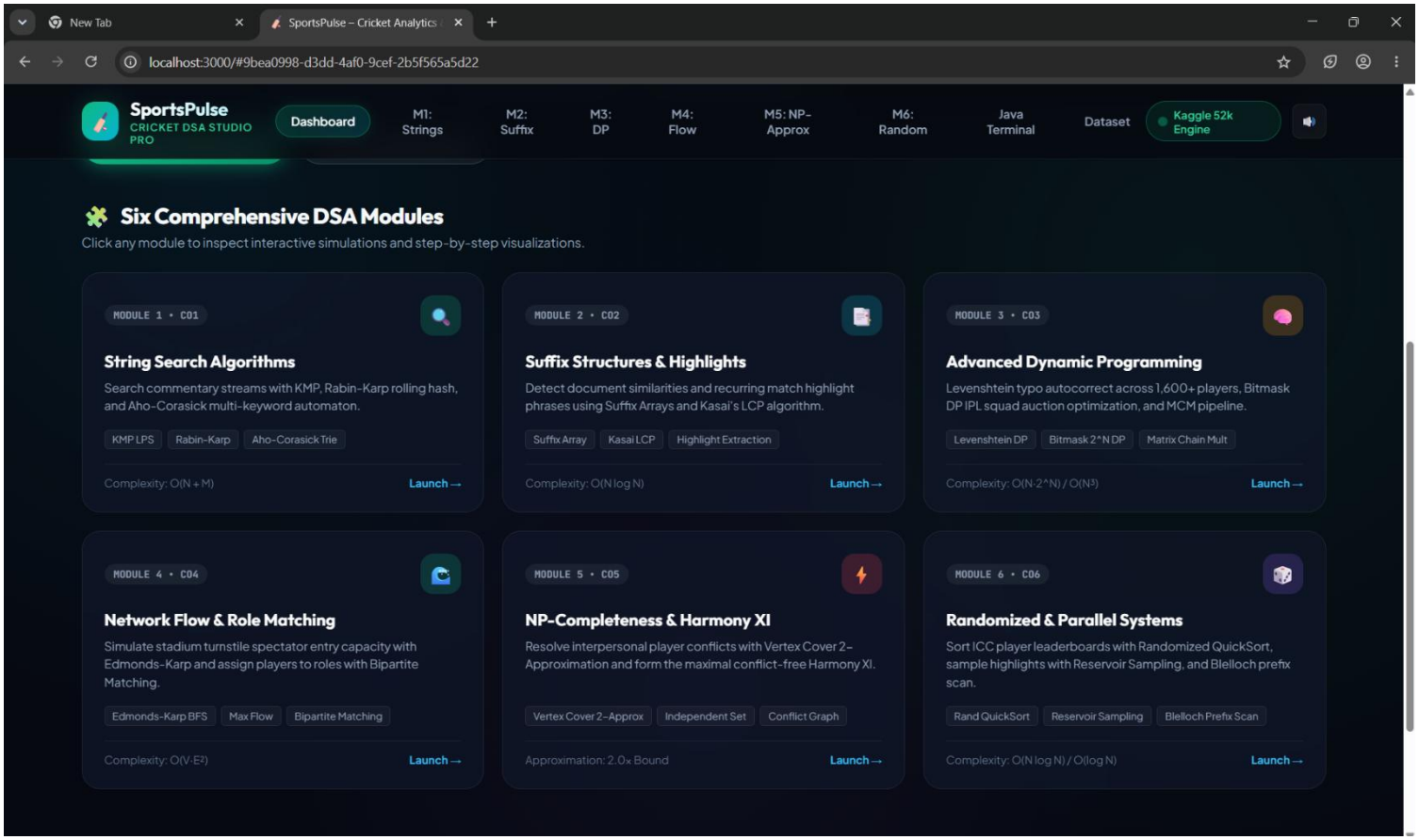
Press Enter to return to the Main Menu...
Indexing completed.
Java: Ready
Ln 1, Col 1
Spaces: 4
UTF-8
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{ } Java
Go Live
Prettier

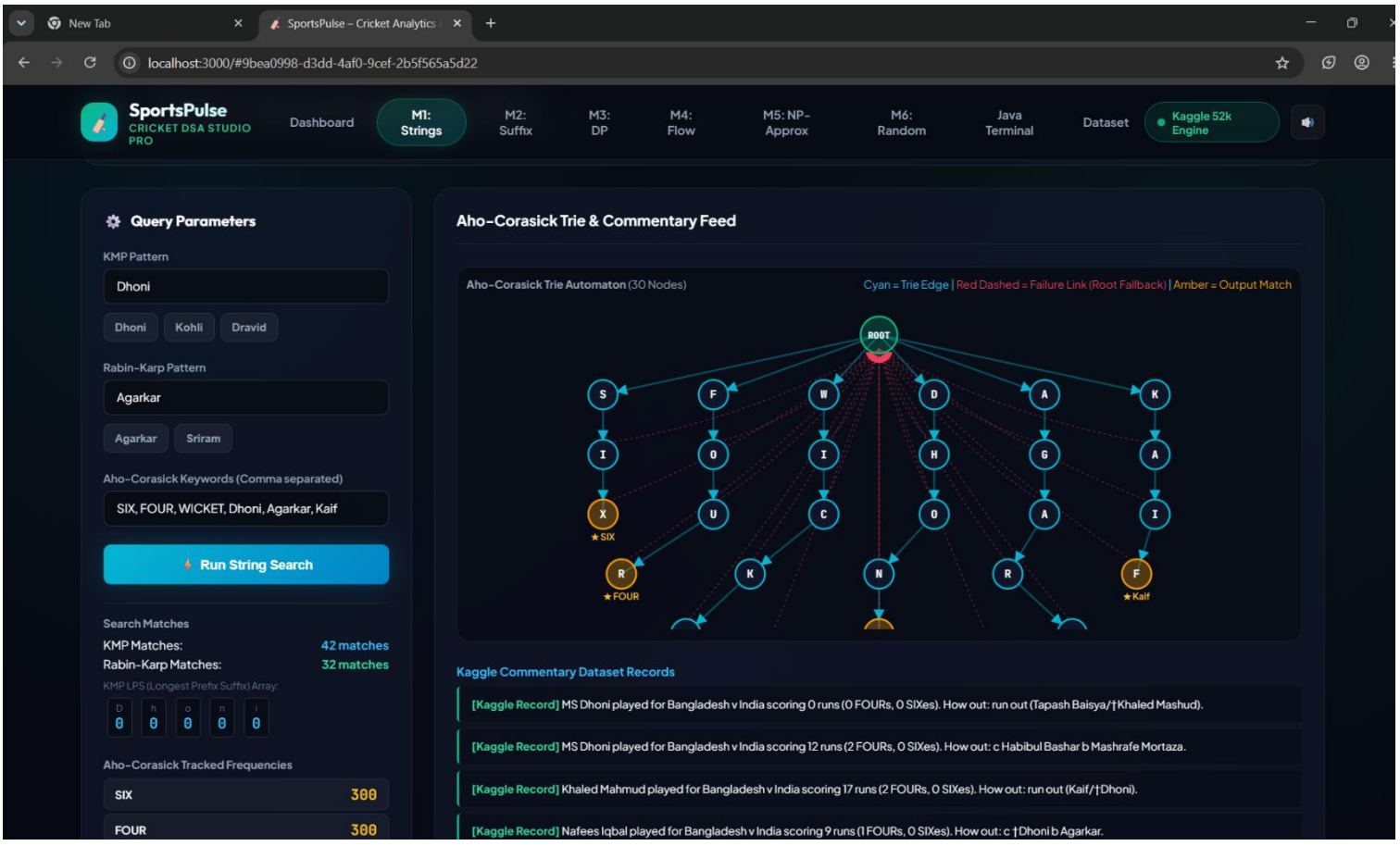

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Problems 33 Output Debug Console Terminal Spell Checker 33
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PS C:\Users\gumud\OneDrive\Desktop\2-1 DSA-3\DSA Sem Project> java SportsPulse
>>
0. Exit
Enter your choice (0-7): 4
MODULE 4: NETWORK FLOW & MATCHING (DYNAMIC)
1. Enter Stadium Gate Capacity (Main Gate -> Security, Press Enter for 150):
Enter Concourse Capacity (Turnstile -> Stands, Press Enter for 100):
[Edmonds-Karp Max Flow Result]: Max Stadium Entry Flow Rate = 180 fans/minute
2. Maximum Bipartite Matching Player-to-Role Assignment:
- Rohit Sharma ---> Powerplay Opener
- Virat Kohli ---> Middle Order Anchor
- Jasprit Bumrah ---> Death Bowler
- Ravindra Jadeja ---> Spin All-Rounder
Press Enter to return to the Main Menu...
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Indexing completed.
Java: Ready
Ln 1, Col 1 Spaces: 4 UTF-8 LF { } Java Go Live Prettier

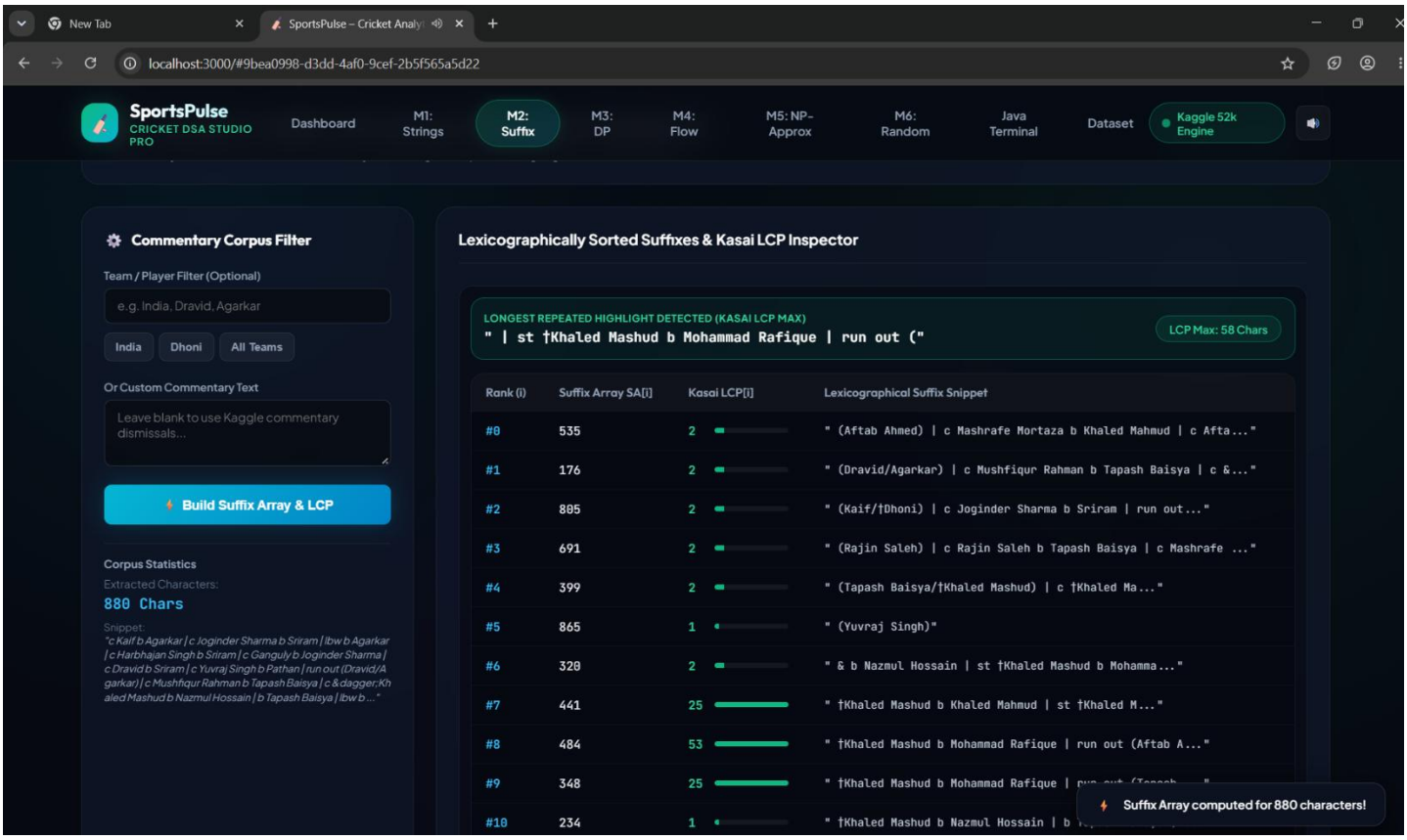

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PS C:\Users\gumud\OneDrive\Desktop\2-1 DSA-3\DSA Sem Project> java SportsPulse
>>
5. Module 5: NP-Approximation Algorithms (Vertex Cover, Independent Set)
6. Module 6: Randomized & Parallel Algorithms (QuickSort, Reservoir, Blelloch)
7. [RUN ALL] Run All 6 Modules Sequentially with Custom/Default Inputs
0. Exit
Enter your choice (0-7): 5
MODULE 5: NP-APPROXIMATION ALGORITHMS (DYNAMIC)
Enter Player Names comma-separated (Press Enter for Default Squad):
>
--- DYNAMIC GRAPH COMPUTATION RESULTS ---
1. Vertex-Cover 2-Approximation Mediator Players: [Kohli, Rohit, Hardik]
2. Maximal Independent Set 100% Harmony Playing XI Roster: [Kohli, Bumrah, Rahul]
Press Enter to return to the Main Menu...
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Indexing completed.
Java: Ready
Ln 1, Col 1 Spaces: 4 UTF-8 LF { } Java Go Live Prettier

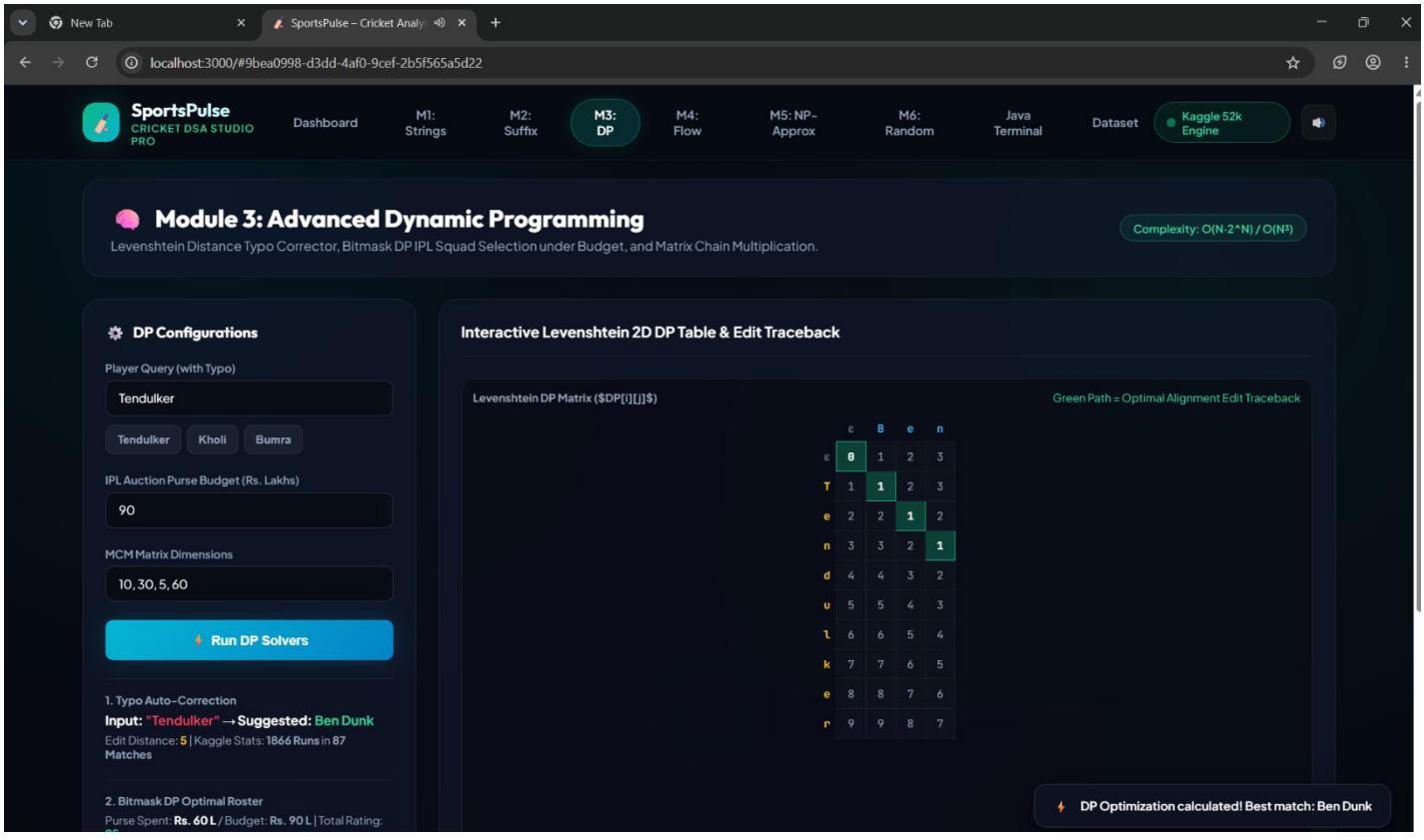

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java
PS C:\Users\gumud\OneDrive\Desktop\2-1 DSA-3\DSA Sem Project> java SportsPulse
>>
==
MODULE 6: RANDOMIZED & PARALLEL ALGORITHMS (DYNAMIC)
==
1. Randomized QuickSort ICC Player Leaderboard Ranking:
Rank #1: Virat Kohli (895.0)
Rank #2: Jasprit Bumrah (887.5)
Rank #3: Suryakumar Yadav (875.0)
Rank #4: MS Dhoni (860.0)
Rank #5: Rohit Sharma (852.0)
2. Reservoir Sampling Highlights. Enter Sample Size k (Press Enter for 3):
Uniformly Sampled 3 Highlight Clips from Stream:
[Clip 1] Over 2.3: SIX by Kohli
[Clip 2] Over 12.2: FOUR through cover
[Clip 3] Over 0.6: 4 Runs by Rohit
3. Enter Runs Scored per 5-Over Phase comma-separated (Press Enter for '45, 38, 52, 60'):
Blelloch Prefix Scan Runs Per Phase : [45, 38, 52, 60]
Cumulative Score Timeline Line : [45, 83, 135, 195]
Press Enter to return to the Main Menu...
Indexing completed
Java: Ready
Ln 1, Col 1
Spaces: 4
UTF-8
LF
{ } Java
Go Live
Prettier

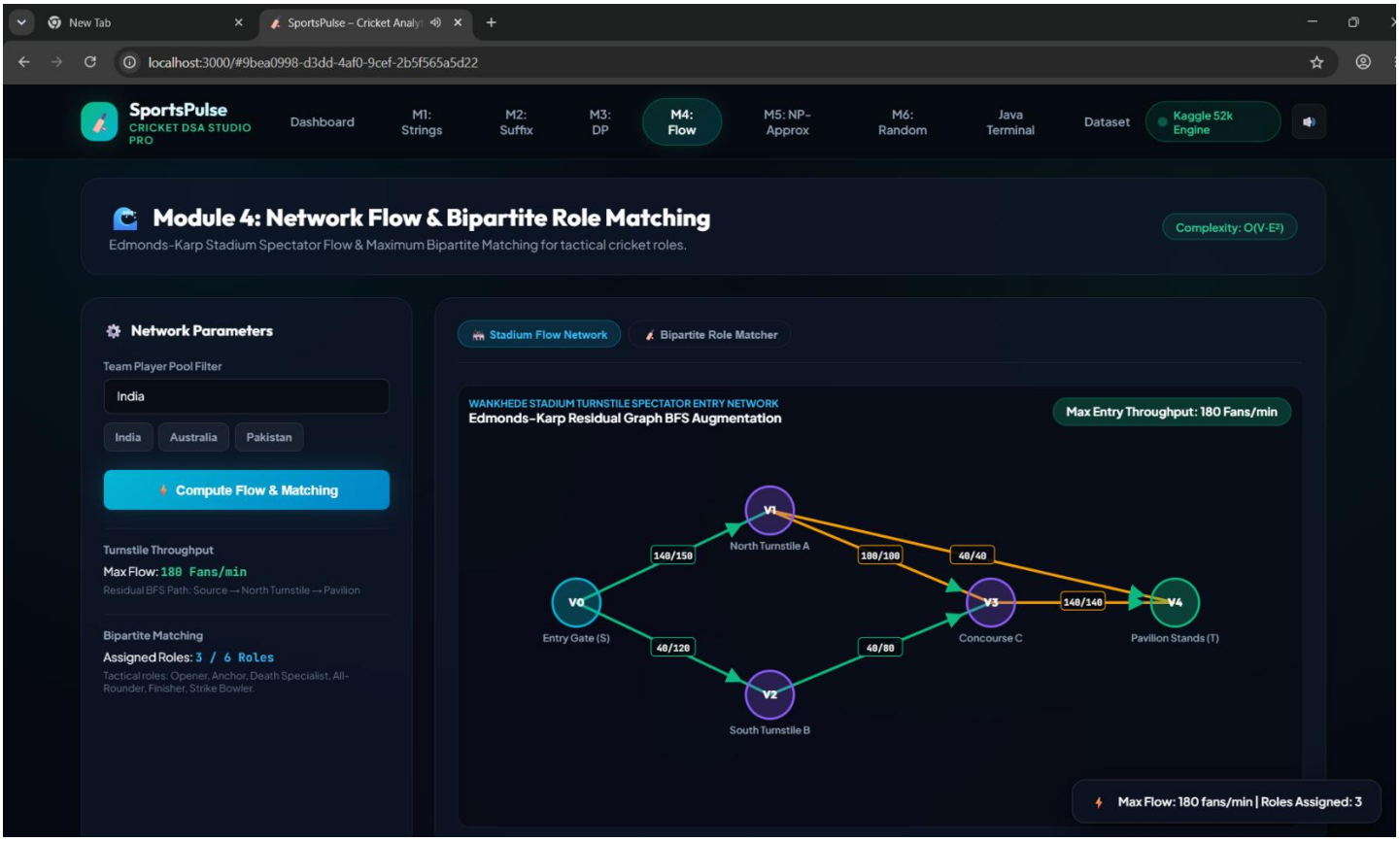

New Tab
SportsPulse – Cricket Analytics
localhost:3000/#9bea0998-d3dd-4af0-9cef-2b5f565a5d22
SportsPulse
CRICKET DSA STUDIO PRO
Dashboard
M1: Strings
M2: Suffix
M3: DP
M4: Flow
M5: NP-Approx
M6: Random
Java Terminal
Dataset
Kaggle 52k Engine
SEMESTER DSA PROJECT • CRICKET PERFORMANCE PLATFORM
Advanced Data Structures & Algorithms in Cricket Analytics
SportsPulse bridges theoretical computer science with real-world sports intelligence. Built over 52,492 Kaggle international cricket ball-by-ball records, featuring 6 core DSA modules, interactive visualizers, and a live Java backend runner.
52,492
Kaggle Ball Records
1,569
International Players
6
Core DSA Modules
100%
Java Binary Parity
HOLOGRAPHIC BROADCAST
HOLOGRAPHIC ARENA
Wankhede Smart Stadium Analytics
Edmonds-Karp 180 Fans/min
Explore Module Studio
Run Java CLI Terminal
Six Comprehensive DSA Modules
Click any module to inspect interactive simulations and step-by-step visualizations.
MODULE 1 • CO1
String Search Algorithms
MODULE 2 • CO2
Suffix Structures & Highlights
MODULE 3 • CO3
Advanced Dynamic Programming

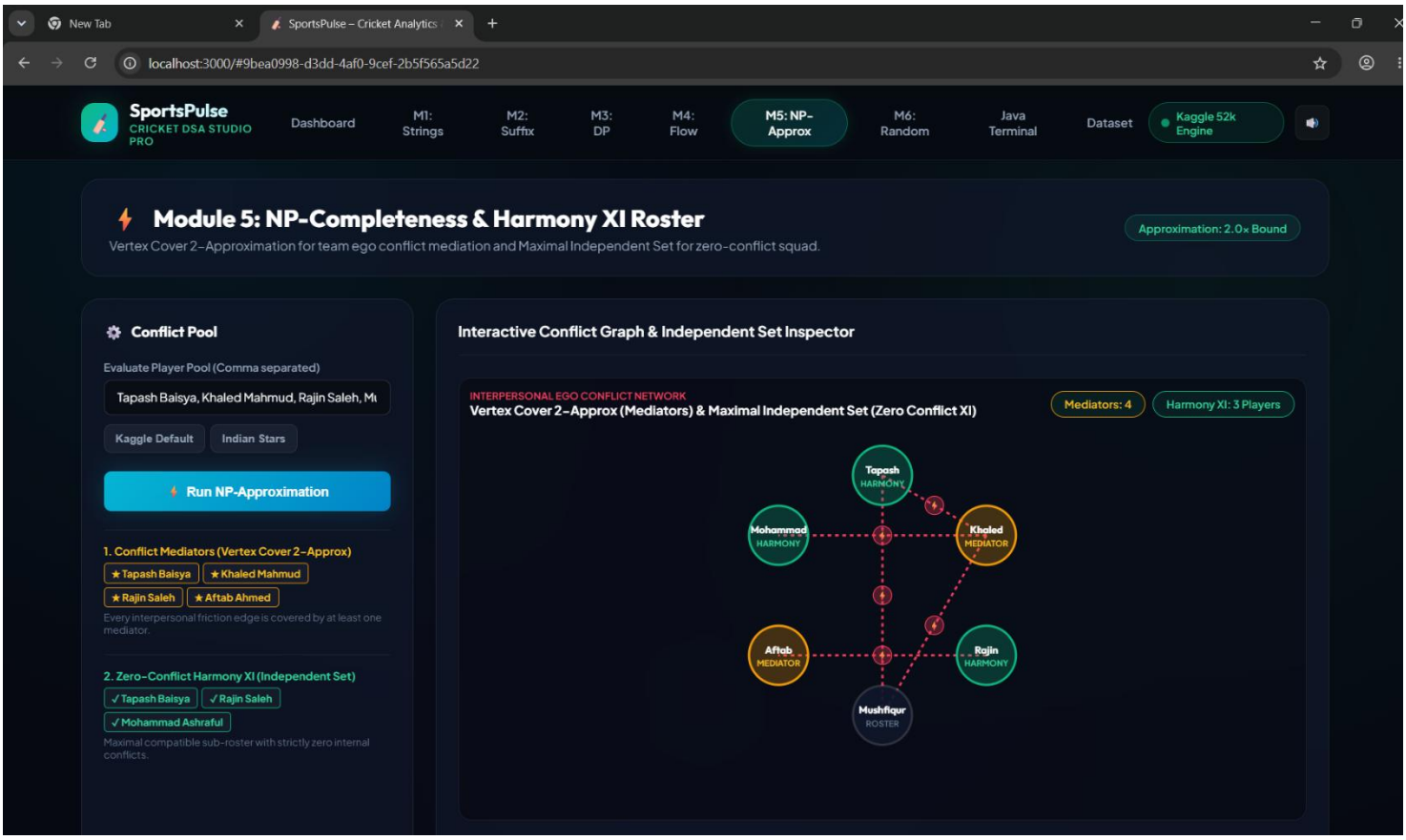

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Java Terminal
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Kaggle 52k Engine
Six Comprehensive DSA Modules
Click any module to inspect interactive simulations and step-by-step visualizations.
MODULE 1 • CO1
String Search Algorithms
Search commentary streams with KMP, Rabin-Karp rolling hash, and Aho-Corasick multi-keyword automaton.
KMP LPS
Rabin-Karp
Aho-Corasick Trie
Complexity: O(N + M)
Launch →
MODULE 2 • CO2
Suffix Structures & Highlights
Detect document similarities and recurring match highlight phrases using Suffix Arrays and Kasai's LCP algorithm.
Suffix Array
Kasai LCP
Highlight Extraction
Complexity: O(N log N)
Launch →
MODULE 3 • CO3
Advanced Dynamic Programming
Levenshtein typo autocorrect across 1,600+ players, Bitmask DP IPL squad auction optimization, and MCM pipeline.
Levenshtein DP
Bitmask 2^N DP
Matrix Chain Mult
Complexity: O(N·2^N) / O(N³)
Launch →
MODULE 4 • CO4
Network Flow & Role Matching
Simulate stadium turnstile spectator entry capacity with Edmonds-Karp and assign players to roles with Bipartite Matching.
Edmonds-Karp BFS
Max Flow
Bipartite Matching
Complexity: O(V·E²)
Launch →
MODULE 5 • CO5
NP-Completeness & Harmony XI
Resolve interpersonal player conflicts with Vertex Cover 2-Approximation and form the maximal conflict-free Harmony XI.
Vertex Cover 2-Approx
Independent Set
Conflict Graph
Approximation: 2.0× Bound
Launch →
MODULE 6 • CO6
Randomized & Parallel Systems
Sort ICC player leaderboards with Randomized QuickSort, sample highlights with Reservoir Sampling, and Blelloch prefix scan.
Rand QuickSort
Reservoir Sampling
Blelloch Prefix Scan
Complexity: O(N log N) / O(log N)
Launch →

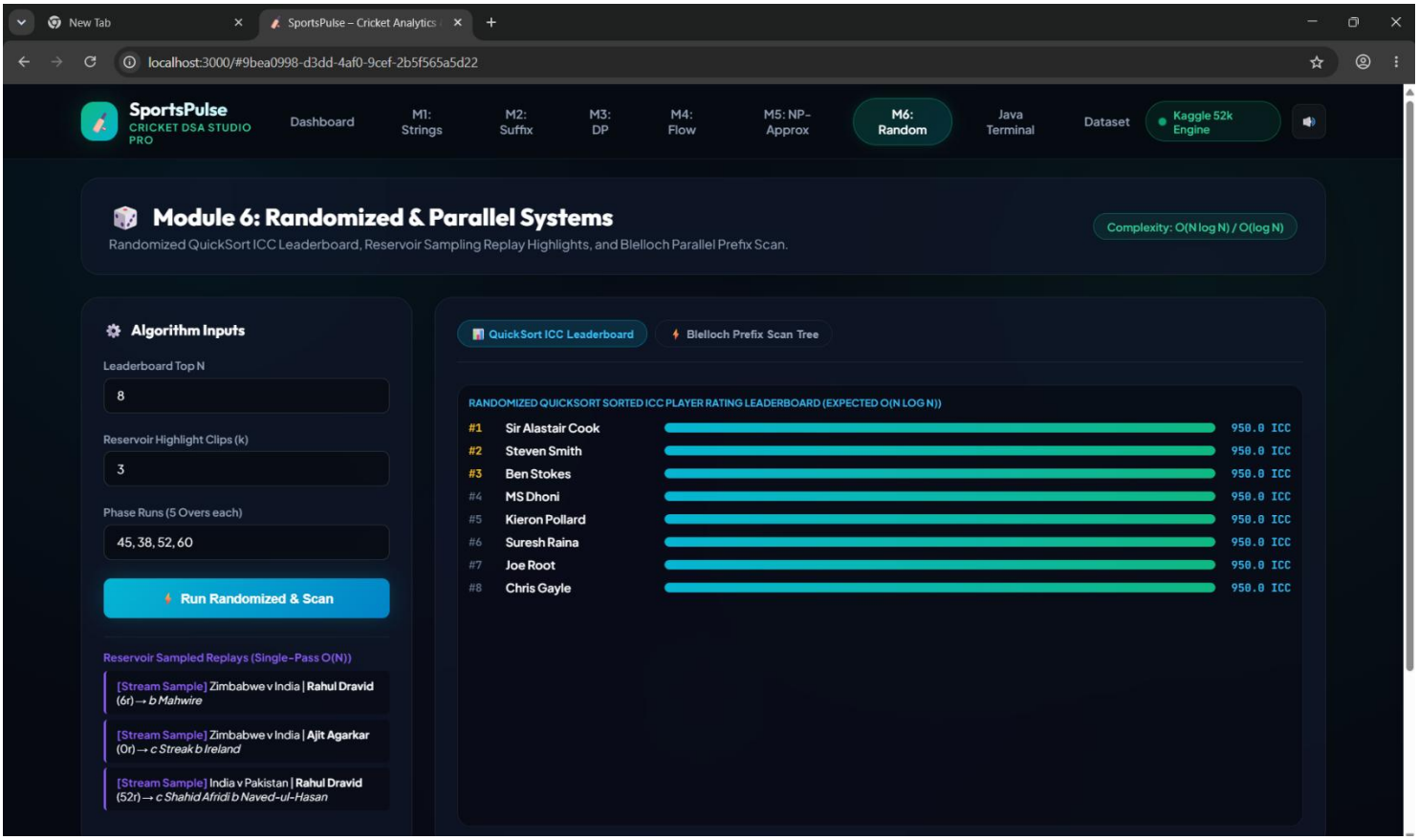

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Dashboard
M1: Strings
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M4: Flow
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M6: Random
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Kaggle 52k Engine
Query Parameters
KMP Pattern
Dhoni
Dhoni
Kohli
Dravid
Rabin-Karp Pattern
Agarkar
Agarkar
Sriram
Aho-Corasick Keywords (Comma separated)
SIX, FOUR, WICKET, Dhoni, Agarkar, Kaif
Run String Search
Search Matches
KMP Matches: 42 matches
Rabin-Karp Matches: 32 matches
KMP LPS (Longest Prefix Suffix) Array:
D 0
h 0
o 0
n 0
i 0
Aho-Corasick Tracked Frequencies
SIX 300
FOUR 300
Aho-Corasick Trie & Commentary Feed
Aho-Corasick Trie Automaton (30 Nodes)
Cyan = Trie Edge | Red Dashed = Failure Link (Root Fallback) | Amber = Output Match
ROOT
★SIX
★FOUR
★Kaif
Kaggle Commentary Dataset Records
[Kaggle Record] MS Dhoni played for Bangladesh v India scoring 0 runs (0 FOURs, 0 SIXes). How out: run out (Tapash Baisya/†Khaled Mashud).
[Kaggle Record] MS Dhoni played for Bangladesh v India scoring 12 runs (2 FOURs, 0 SIXes). How out: c Habibul Bashar b Mashrafe Mortaza.
[Kaggle Record] Khaled Mahmud played for Bangladesh v India scoring 17 runs (2 FOURs, 0 SIXes). How out: run out (Kaif/†Dhoni).
[Kaggle Record] Nafees Iqbal played for Bangladesh v India scoring 9 runs (1 FOURs, 0 SIXes). How out: c †Dhoni b Agarkar.

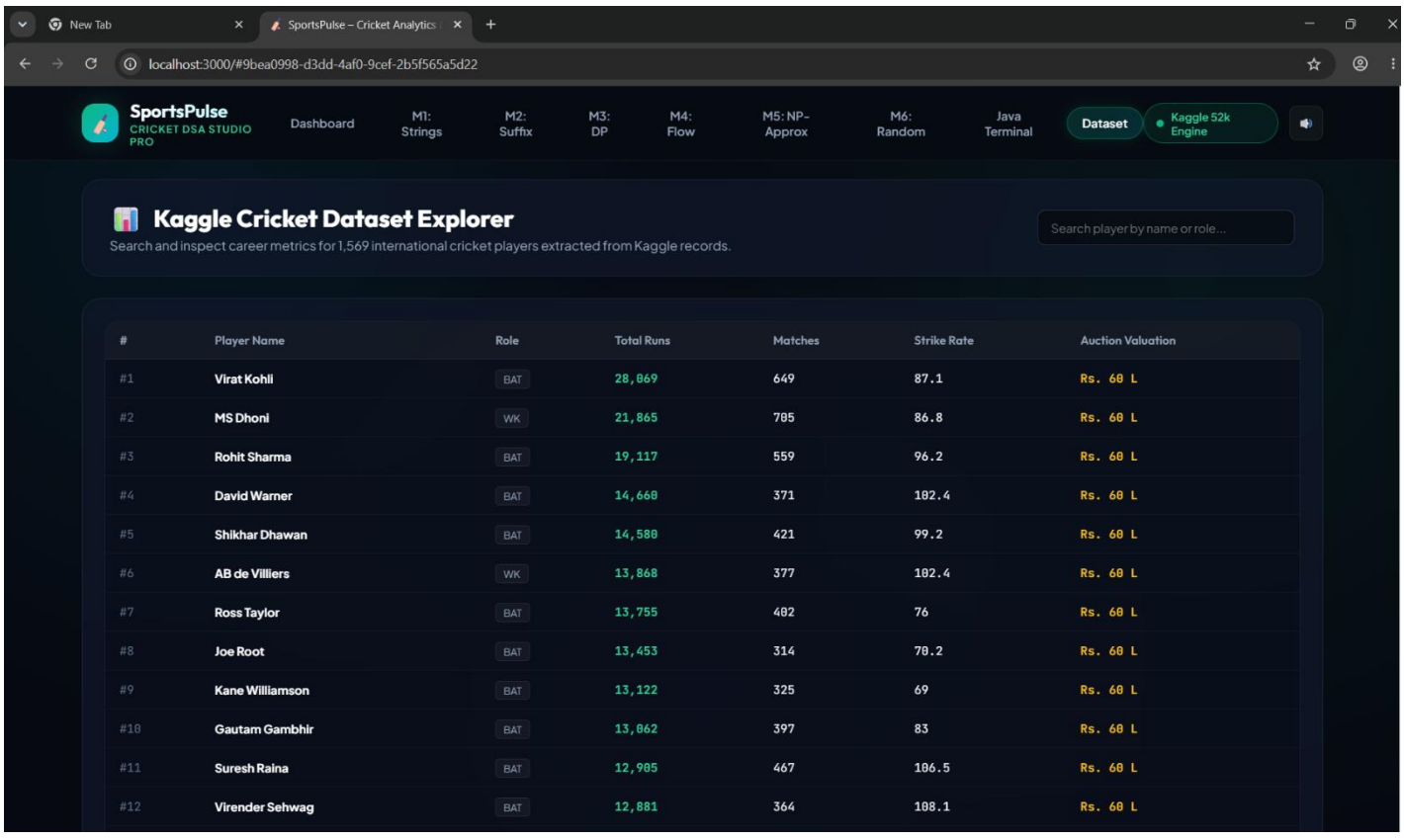

New Tab
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SportsPulse
CRICKET DSA STUDIO PRO
Dashboard
M1: Strings
M2: Suffix
M3: DP
M4: Flow
M5: NP-Approx
M6: Random
Java Terminal
Dataset
Kaggle 52k Engine
Commentary Corpus Filter
Team / Player Filter (Optional)
e.g. India, Dravid, Agarkar
India
Dhoni
All Teams
Or Custom Commentary Text
Leave blank to use Kaggle commentary dismissals...
Build Suffix Array & LCP
Corpus Statistics
Extracted Characters:
880 Chars
Snippet:
"c Kaif b Agarkar | c Joginder Sharma b Sriram | lbw b Agarkar | c Harbhajan Singh b Sriram | c Ganguly b Joginder Sharma | c Dravid b Sriram | c Yuvraj Singh b Pathan | run out (Dravid/Agarkar) | c Mushfiqur Rahman b Tapash Baisya | c †Khaled Mashud b Nazmul Hossain | b Tapash Baisya | lbw b ..."
Lexicographically Sorted Suffixes & Kasai LCP Inspector
LONGEST REPEATED HIGHLIGHT DETECTED (KASAI LCP MAX)
" | st †Khaled Mashud b Mohammad Rafique | run out ("
LCP Max: 58 Chars
Rank (i)
Suffix Array SA[i]
Kasai LCP[i]
Lexicographical Suffix Snippet
#0 535 2 " (Aftab Ahmed) | c Mashrafe Mortaza b Khaled Mahmud | c Afta..."
#1 176 2 " (Dravid/Agarkar) | c Mushfiqur Rahman b Tapash Baisya | c &..."
#2 885 2 " (Kaif/†Dhoni) | c Joginder Sharma b Sriram | run out..."
#3 691 2 " (Rajin Saleh) | c Rajin Saleh b Tapash Baisya | c Mashrafe ..."
#4 399 2 " (Tapash Baisya/†Khaled Mashud) | c †Khaled Ma..."
#5 865 1 " (Yuvraj Singh)"
#6 320 2 " & b Nazmul Hossain | st †Khaled Mashud b Mohamma..."
#7 441 25 " †Khaled Mashud b Khaled Mahmud | st †Khaled M..."
#8 484 53 " †Khaled Mashud b Mohammad Rafique | run out (Aftab A..."
#9 348 25 " †Khaled Mashud b Mohammad Rafique |
#10 234 1 " †Khaled Mashud b Nazmul Hossain | b
Suffix Array computed for 880 characters!


New Tab
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Kaggle 52k Engine
Module 3: Advanced Dynamic Programming
Levenshtein Distance Typo Corrector, Bitmask DP IPL Squad Selection under Budget, and Matrix Chain Multiplication.
Complexity: O(N·2^N) / O(N³)
DP Configurations
Player Query (with Typo)
Tendulker
Tendulker
Kholi
Bumra
IPL Auction Purse Budget (Rs. Lakhs)
90
MCM Matrix Dimensions
10, 30, 5, 60
Run DP Solvers
1. Typo Auto-Correction
Input: "Tendulker" → Suggested: Ben Dunk
Edit Distance: 5 | Kaggle Stats: 1866 Runs in 87 Matches
2. Bitmask DP Optimal Roster
Purse Spent: Rs. 60 L / Budget: Rs. 90 L | Total Rating:
Interactive Levenshtein 2D DP Table & Edit Traceback
Levenshtein DP Matrix ($DP[i][j]$)
Green Path = Optimal Alignment Edit Traceback
ε B e n
ε 0 1 2 3
T 1 1 2 3
e 2 2 1 2
n 3 3 2 1
d 4 4 3 2
u 5 5 4 3
l 6 6 5 4
k 7 7 6 5
e 8 8 7 6
r 9 9 8 7
DP Optimization calculated! Best match: Ben Dunk


New Tab
SportsPulse – Cricket Analy
localhost:3000/#9bea0998-d3dd-4af0-9cef-2b5f565a5d22
SportsPulse
CRICKET DSA STUDIO PRO
Dashboard
M1: Strings
M2: Suffix
M3: DP
M4: Flow
M5: NP-Approx
M6: Random
Java Terminal
Dataset
Kaggle 52k Engine
Module 4: Network Flow & Bipartite Role Matching
Edmonds-Karp Stadium Spectator Flow & Maximum Bipartite Matching for tactical cricket roles.
Complexity: O(V·E²)
Network Parameters
Team Player Pool Filter
India
India
Australia
Pakistan
Compute Flow & Matching
Turnstile Throughput
Max Flow: 180 Fans/min
Residual BFS Path: Source → North Turnstile → Pavilion
Bipartite Matching
Assigned Roles: 3 / 6 Roles
Tactical roles: Opener, Anchor, Death Specialist, All-Rounder, Finisher, Strike Bowler.
Stadium Flow Network
Bipartite Role Matcher
WANKHEDE STADIUM TURNSTILE SPECTATOR ENTRY NETWORK
Edmonds-Karp Residual Graph BFS Augmentation
Max Entry Throughput: 180 Fans/min
V0
Entry Gate (S)
V1
North Turnstile A
V2
South Turnstile B
V3
Concourse C
V4
Pavilion Stands (T)
140/150
100/100
40/40
140/140
40/120
40/80
Max Flow: 180 fans/min | Roles Assigned: 3


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Module 5: NP-Completeness & Harmony XI Roster
Vertex Cover 2-Approximation for team ego conflict mediation and Maximal Independent Set for zero-conflict squad.
Approximation: 2.0× Bound
Conflict Pool
Evaluate Player Pool (Comma separated)
Tapash Baisya, Khaled Mahmud, Rajin Saleh, Mı
Kaggle Default
Indian Stars
Run NP-Approximation
1. Conflict Mediators (Vertex Cover 2-Approx)
★ Tapash Baisya
★ Khaled Mahmud
★ Rajin Saleh
★ Aftab Ahmed
Every interpersonal friction edge is covered by at least one mediator.
2. Zero-Conflict Harmony XI (Independent Set)
✓ Tapash Baisya
✓ Rajin Saleh
✓ Mohammad Ashraful
Maximal compatible sub-roster with strictly zero internal conflicts.
Interactive Conflict Graph & Independent Set Inspector
INTERPERSONAL EGO CONFLICT NETWORK
Vertex Cover 2-Approx (Mediators) & Maximal Independent Set (Zero Conflict XI)
Mediators: 4
Harmony XI: 3 Players
Tapash HARMONY
Mohammad HARMONY
Khaled MEDIATOR
Aftab MEDIATOR
Rajin HARMONY
Mushfiqur ROSTER


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Module 6: Randomized & Parallel Systems
Randomized QuickSort ICC Leaderboard, Reservoir Sampling Replay Highlights, and Blelloch Parallel Prefix Scan.
Complexity: O(N log N) / O(log N)
Algorithm Inputs
Leaderboard Top N
8
Reservoir Highlight Clips (k)
3
Phase Runs (5 Overs each)
45, 38, 52, 60
Run Randomized & Scan
Reservoir Sampled Replays (Single-Pass O(N))
[Stream Sample] Zimbabwe v India | Rahul Dravid (6r) → b Mahwire
[Stream Sample] Zimbabwe v India | Ajit Agarkar (0r) → c Streak b Ireland
[Stream Sample] India v Pakistan | Rahul Dravid (52r) → c Shahid Afridi b Naved-ul-Hasan
QuickSort ICC Leaderboard
Blelloch Prefix Scan Tree
RANDOMIZED QUICKSORT SORTED ICC PLAYER RATING LEADERBOARD (EXPECTED O(N LOG N))
#1 Sir Alastair Cook 950.0 ICC
#2 Steven Smith 950.0 ICC
#3 Ben Stokes 950.0 ICC
#4 MS Dhoni 950.0 ICC
#5 Kieron Pollard 950.0 ICC
#6 Suresh Raina 950.0 ICC
#7 Joe Root 950.0 ICC
#8 Chris Gayle 950.0 ICC


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Kaggle 52k Engine
Kaggle Cricket Dataset Explorer
Search and inspect career metrics for 1,569 international cricket players extracted from Kaggle records.
Search player by name or role...
Player Name Role Total Runs Matches Strike Rate Auction Valuation
#1 Virat Kohli BAT 28,069 649 87.1 Rs. 60 L
#2 MS Dhoni WK 21,865 705 86.8 Rs. 60 L
#3 Rohit Sharma BAT 19,117 559 96.2 Rs. 60 L
#4 David Warner BAT 14,660 371 102.4 Rs. 60 L
#5 Shikhar Dhawan BAT 14,580 421 99.2 Rs. 60 L
#6 AB de Villiers WK 13,868 377 102.4 Rs. 60 L
#7 Ross Taylor BAT 13,755 402 76 Rs. 60 L
#8 Joe Root BAT 13,453 314 70.2 Rs. 60 L
#9 Kane Williamson BAT 13,122 325 69 Rs. 60 L
#10 Gautam Gambhir BAT 13,062 397 83 Rs. 60 L
#11 Suresh Raina BAT 12,905 467 106.5 Rs. 60 L
#12 Virender Sehwag BAT 12,881 364 108.1 Rs. 60 L